CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

ALVES & SON DAIRY #1

13745 NEWPORT RD

TURLOCK CA 95380

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe Alves & Son Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
13745 Newport RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1978

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X041-X110-X014-XXXX

B. OPERATORS

Alves, Joe

Operator name: Alves, Joe

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
13745 Newport RD	Ballico	CA	95303

This operator is responsible for paying permit fees.

C. OWNERS

Alves, Joe

Legal owner name: Alves, Joe

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
13745 Newport RD	Ballico	CA	95303

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	85	218	200	97	130
Number under roof	773	0	0	0	0	0
Maximum number	773	85	218	200	97	130
Average number	773	85	218	200	97	130
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 61 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,346.60 *tons per reporting period*

Total nitrogen from manure: 315,353.98 *lbs per reporting period*

After ammonia losses (30% loss applied): 220,747.79 *lbs per reporting period*

Total phosphorus from manure: 51,507.38 *lbs per reporting period*

Total potassium from manure: 143,465.88 *lbs per reporting period*

Total salt from manure: 383,512.80 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 6,547,500 *gallons*

Total nitrogen generated: 33,084.18 *lbs*

Total phosphorus generated: 5,071.05 *lbs*

Total potassium generated: 42,605.94 *lbs*

Total salt generated: 172,619.45 *lbs*

	6,547,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	6,547,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	53	50	3	process wastewater	X041-X110-X012-XXXX X041-X110-X013-XXXX
2	35	35	3	process wastewater	X041-X110-X014-XXXX
3	99	30	3	process wastewater	X041-X110-X014-XXXX
6	134	134	1	none	X042-X030-X003-XXXX
Totals for areas that were used for application	187	115	9		
Totals for areas that were not used for application	134	134	1		
Land application area totals	321	249	10		

B. CROPS AND HARVESTS

1

Field name: 1

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	715.00 *ton*	Dry-weight		66.2	20,200.00	2,900.00	32,900.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.30	195.27	28.03	318.04	947.35

1

05/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 50 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/26/2023	327.00 *ton*	Dry-weight		65.8	23,500.00	3,800.00	37,000.00		11.60
09/25/2023	210.00 *ton*	Dry-weight		54.1	24,500.00	3,300.00	39,600.00		15.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.74	199.59	29.72	318.20	1,128.09

2

Field name: 2

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	501.00 *ton*	Dry-weight		78.2	11,000.00	2,400.00	22,100.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.31	68.65	14.98	137.93	443.11

2

05/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 35 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/26/2023	229.00 *ton*	Dry-weight		66.2	23,800.00	3,700.00	37,200.00		11.80
09/25/2023	147.00 *ton*	Dry-weight		66.3	22,900.00	3,200.00	39,000.00		16.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.74	170.09	25.42	274.94	977.67

3

Field name: 3

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	429.00 *ton*	Dry-weight		74.3	20,400.00	3,300.00	26,100.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.30	149.94	24.26	191.84	661.52

3

05/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 30 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/26/2023	196.00 *ton*	Dry-weight		16.7	19,900.00	3,100.00	42,100.00		13.20
09/25/2023	126.00 *ton*	Dry-weight		77.0	24,500.00	4,000.00	43,500.00		17.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.73	263.94	41.47	542.28	1,767.13

6

Field name: 6

01/01/2015: Almond, in shell

Crop: Almond, in shell Acres planted: 134 Plant date: 01/01/2015

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	62.00 *ton*	Dry-weight		6.1	22,900.00	2,400.00	13,000.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.50	195.00	30.00	150.00	0.00
Total actual harvest content	0.46	19.90	2.09	11.30	90.37

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/03/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.94	15.58	136.20	549.21	1,012,500.00 *gal*
Irrigation Well	Ground water	14.60	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		120.54	15.58	136.20	549.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	8,100,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.94	15.58	136.20	549.21	1,012,500.00 *gal*
Irrigation Well	Ground water	17.52	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		123.46	15.58	136.20	549.21	

1 - 05/20/2023: Sudangrass, silage

Field name: 1

Crop: Sudangrass, silage Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 05/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.67	5.54	48.43	195.27	360,000.00 *gal*
Irrigation Well	Ground water	21.17	0.00	0.00	0.00	9,396,000.00 *gal*
Application event totals		58.84	5.54	48.43	195.27	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.50	8.31	72.64	292.91	540,000.00 *gal*
Irrigation Well	Ground water	17.28	0.00	0.00	0.00	7,668,000.00 *gal*
Application event totals		73.78	8.31	72.64	292.91	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.76	0.00	0.00	0.00	7,884,000.00 *gal*
Application event totals		17.76	0.00	0.00	0.00	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.12	8.98	43.98	193.77	540,000.00 *gal*
Irrigation Well	Ground water	21.17	0.00	0.00	0.00	9,396,000.00 *gal*
Application event totals		54.29	8.98	43.98	193.77	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.03	0.00	0.00	0.00	7,560,000.00 *gal*
Application event totals		17.03	0.00	0.00	0.00	

1 - 05/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.49	0.00	0.00	0.00	8,208,000.00 *gal*
Application event totals		18.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	21.41	0.00	0.00	0.00	9,504,000.00 *gal*
Application event totals		21.41	0.00	0.00	0.00	

2 - 10/03/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.36	5.94	51.89	209.22	270,000.00 *gal*
Irrigation Well	Ground water	14.60	0.00	0.00	0.00	4,536,000.00 *gal*
Application event totals		54.96	5.94	51.89	209.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	5,670,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.56	0.00	0.00	0.00	5,454,000.00 *gal*
Application event totals		17.56	0.00	0.00	0.00	

2 - 05/20/2023: Sudangrass, silage

Field name: 2

Crop: Sudangrass, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.08	6.92	60.53	244.09	315,000.00 *gal*
Irrigation Well	Ground water	21.12	0.00	0.00	0.00	6,561,000.00 *gal*
Application event totals		68.20	6.92	60.53	244.09	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.08	6.92	60.53	244.09	315,000.00 *gal*
Irrigation Well	Ground water	17.29	0.00	0.00	0.00	5,373,000.00 *gal*
Application event totals		64.38	6.92	60.53	244.09	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.90	0.00	0.00	0.00	5,562,000.00 *gal*
Application event totals		17.90	0.00	0.00	0.00	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	21.12	0.00	0.00	0.00	6,561,000.00 *gal*
Application event totals		21.12	0.00	0.00	0.00	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.03	0.00	0.00	0.00	5,292,000.00 *gal*
Application event totals		17.03	0.00	0.00	0.00	

2 - 05/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.42	0.00	0.00	0.00	5,724,000.00 *gal*
Application event totals		18.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	21.55	0.00	0.00	0.00	6,696,000.00 *gal*
Application event totals		21.55	0.00	0.00	0.00	

3 - 10/03/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.12	11.19	97.86	394.61	436,500.00 *gal*
Irrigation Well	Ground water	14.60	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		90.72	11.19	97.86	394.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	4,860,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

3 - 10/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.47	11.54	100.89	406.82	450,000.00 *gal*
Irrigation Well	Ground water	17.64	0.00	0.00	0.00	4,698,000.00 *gal*
Application event totals		96.11	11.54	100.89	406.82	

3 - 05/20/2023: Sudangrass, silage

Field name: 3

Crop: Sudangrass, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.01	7.50	65.58	264.43	292,500.00 *gal*
Irrigation Well	Ground water	21.09	0.00	0.00	0.00	5,616,000.00 *gal*
Application event totals		72.10	7.50	65.58	264.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.01	7.50	65.58	264.43	292,500.00 *gal*
Irrigation Well	Ground water	17.44	0.00	0.00	0.00	4,644,000.00 *gal*
Application event totals		68.45	7.50	65.58	264.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.84	0.00	0.00	0.00	4,752,000.00 *gal*
Application event totals		17.84	0.00	0.00	0.00	

3 - 05/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.99	9.12	79.70	321.39	355,500.00 *gal*
Irrigation Well	Ground water	21.09	0.00	0.00	0.00	5,616,000.00 *gal*
Application event totals		83.08	9.12	79.70	321.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.03	0.00	0.00	0.00	4,536,000.00 *gal*
Application event totals		17.03	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.99	9.12	79.70	321.39	355,500.00 *gal*
Irrigation Well	Ground water	18.66	0.00	0.00	0.00	4,968,000.00 *gal*
Application event totals		80.65	9.12	79.70	321.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	21.50	0.00	0.00	0.00	5,724,000.00 *gal*
Application event totals		21.50	0.00	0.00	0.00	

6 - 01/01/2015: Almond, in shell

Field name: 6

Crop: Almond, in shell

Plant date: 01/01/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

6 - 01/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

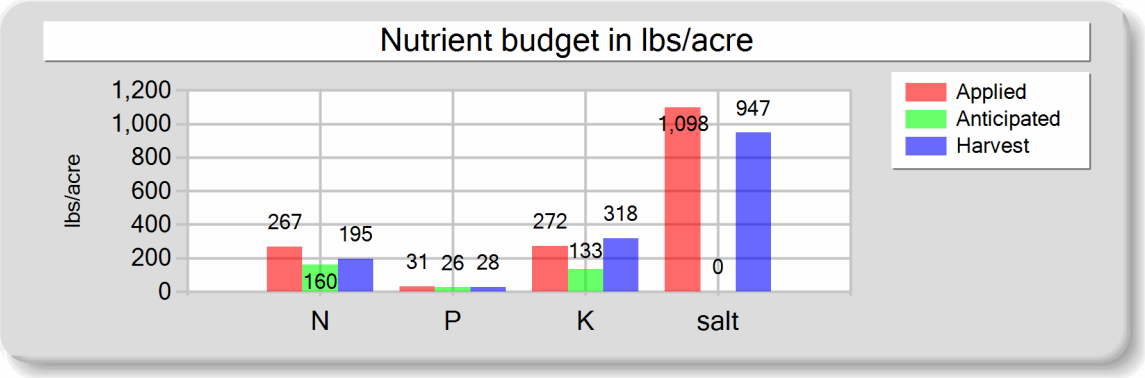
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	18.25	0.00	0.00	0.00	21,708,000.00 *gal*
Application event totals		18.25	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 10/03/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.87	31.16	272.41	1,098.41
Fresh water	50.37	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	266.91	31.16	272.41	1,098.41
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	195.27	28.03	318.04	947.35
Nutrient balance	71.64	3.13	-45.63	151.06
Applied to removed ratio	1.37	1.11	0.86	1.16

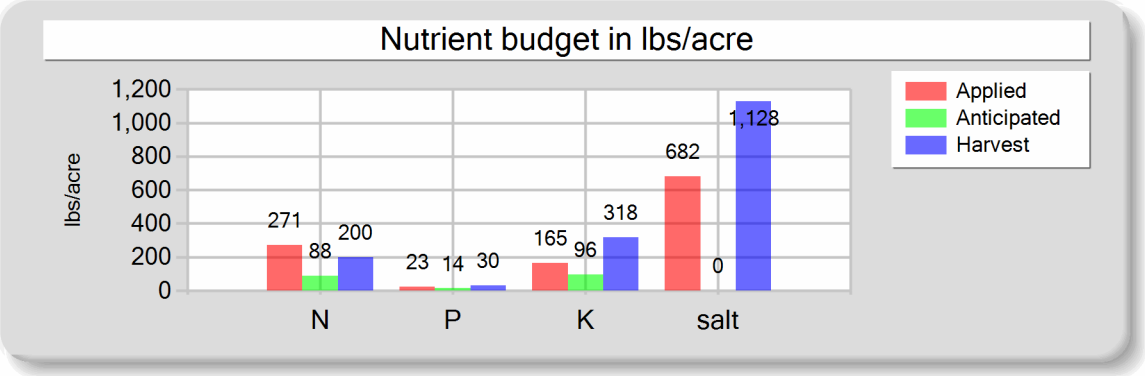
Fresh water applied	
22,356,000.00	*gallons*
823.30	*acre-inches*
16.47	*inches/acre*

Process wastewater applied	
2,025,000.00	*gallons*
74.57	*acre-inches*
1.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/20/2023: Sudangrass, silage

Field name: 1 Crop: Sudangrass, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	127.29	22.83	165.05	681.95
Fresh water	134.32	0.00	0.00	0.00
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	270.95	22.83	165.05	681.95
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	199.59	29.72	318.20	1,128.09
Nutrient balance	71.36	-6.90	-153.15	-446.14
Applied to removed ratio	1.36	0.77	0.52	0.60

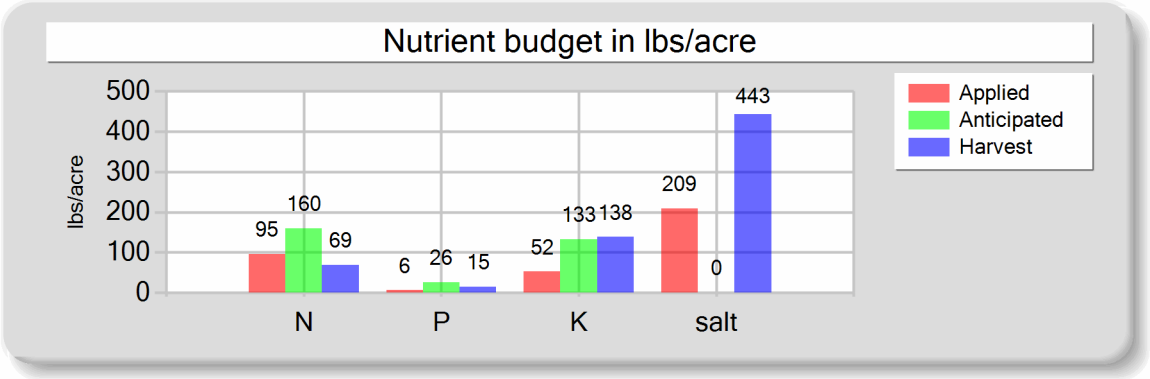
Fresh water applied	
59,616,000.00	*gallons*
2,195.45	*acre-inches*
43.91	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
2	*harvests*

2 - 10/03/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	40.36	5.94	51.89	209.22
Fresh water	50.41	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	95.43	5.94	51.89	209.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	68.65	14.98	137.93	443.11
Nutrient balance	26.78	-9.04	-86.04	-233.89
Applied to removed ratio	1.39	0.40	0.38	0.47

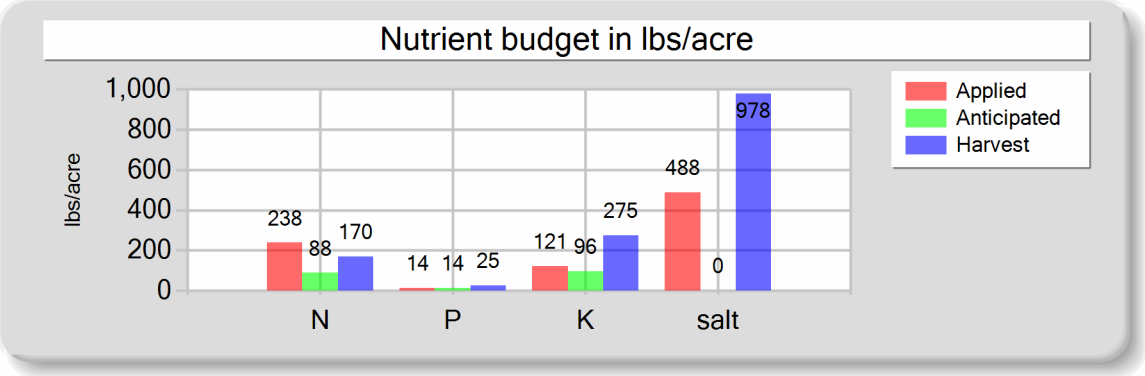
Fresh water applied	
15,660,000.00	*gallons*
576.70	*acre-inches*
16.48	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/20/2023: Sudangrass, silage

Field name: 2 Crop: Sudangrass, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.17	13.85	121.07	488.18
Fresh water	134.45	0.00	0.00	0.00
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	237.95	13.85	121.07	488.18
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	170.09	25.42	274.94	977.67
Nutrient balance	67.85	-11.57	-153.87	-489.49
Applied to removed ratio	1.40	0.54	0.44	0.50

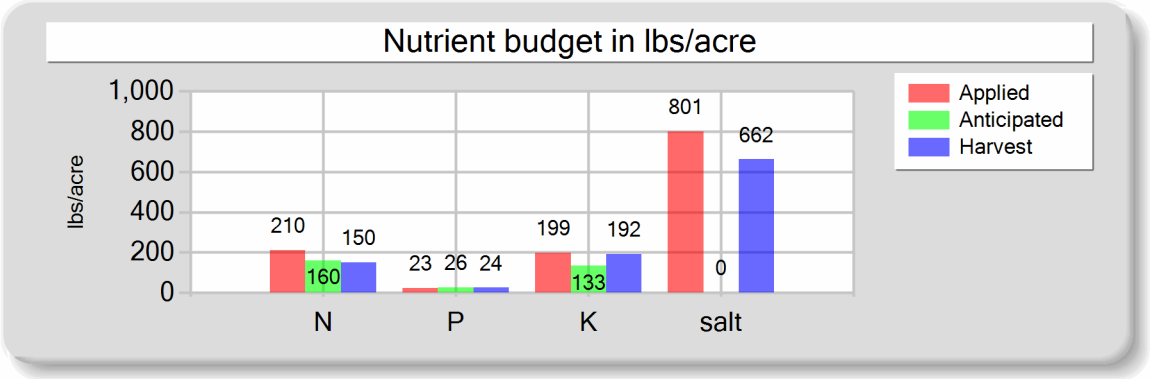
Fresh water applied	
41,769,000.00	*gallons*
1,538.21	*acre-inches*
43.95	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
0.66	*inches/acre*

Total harvests for the crop	
2	*harvests*

3 - 10/03/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.59	22.74	198.76	801.43
Fresh water	50.49	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	209.75	22.74	198.76	801.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	149.94	24.26	191.84	661.52
Nutrient balance	59.81	-1.52	6.92	139.91
Applied to removed ratio	1.40	0.94	1.04	1.21

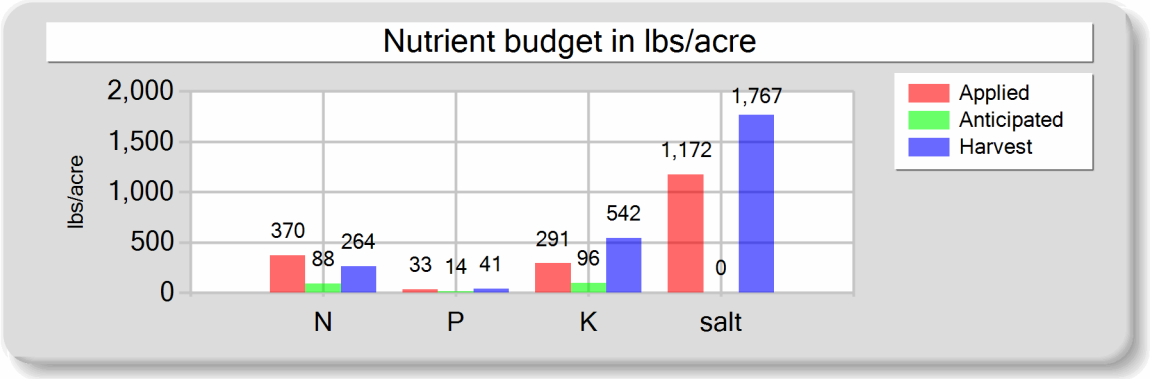
Fresh water applied	
13,446,000.00	*gallons*
495.17	*acre-inches*
16.51	*inches/acre*

Process wastewater applied	
886,500.00	*gallons*
32.65	*acre-inches*
1.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/20/2023: Sudangrass, silage

Field name: 3 Crop: Sudangrass, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.00	33.24	290.57	1,171.64
Fresh water	134.65	0.00	0.00	0.00
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	369.98	33.24	290.57	1,171.64
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	263.94	41.47	542.28	1,767.13
Nutrient balance	106.05	-8.23	-251.71	-595.49
Applied to removed ratio	1.40	0.80	0.54	0.66

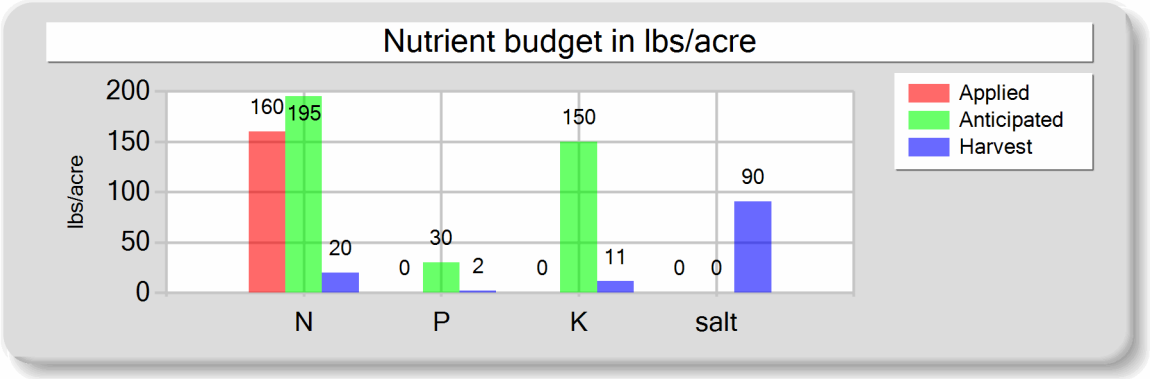
Fresh water applied	
35,856,000.00	*gallons*
1,320.45	*acre-inches*
44.02	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
2	*harvests*

6 - 01/01/2015: Almond, in shell

Field name: 6 Crop: Almond, in shell Plant date: 01/01/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	146.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	160.00	0.00	0.00	0.00
Anticipated crop nutrient removal	195.00	30.00	150.00	0.00
Actual crop nutrient removal	19.90	2.09	11.30	90.37
Nutrient balance	140.11	-2.09	-11.30	-90.37
Applied to removed ratio	8.04	0.00	0.00	0.00

Fresh water applied	
173,664,000.00	*gallons*
6,395.45	*acre-inches*
47.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	2,100.00	12,300.00							0.00
DL	0.01	0.01	0.01							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 11/09/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	2,900.00	6,100.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/02/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,180.00	11.40	0.00	1.60	436.00	859.00								7,200.00	2,150
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,450.00	377.00	0.00	0.70	382.00	972.00								5,870.00	2,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	626.00	428.00	0.00	0.90	92.20	806.00								7,560.00	3,250
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	367.00	338.00	0.00	0.50	99.60	488.00								5,410.00	2,150
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	210.00	179.00	0.00	0.00	44.70	299.00								3,090.00	1,670
DL	0.70	0.20	0.01	0.20	0.02	0.02								2.00	10

C. FRESH WATER ANALYSES

Domestic Well

Domestic Well Milk Barn

Sample description: Domestic Well Milk Barn

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.70	0.00	8.70								962.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well Silage

Sample description: Domestic Well Silage

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.90	0.00	3.90								235.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

Sample description: Irrigation Well

Sample date: 11/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.50	0.00	13.50								837.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/03/2022: Oats, silage-soft dough

Field 1 Oat Silage

Sample and source description: Field 1 Oat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	2,900.00	32,900.00		9.80
DL	0.01	0.01	0.01		1.00

1 - 05/20/2023: Sudangrass, silage

Field 1 Sudan Silage

Sample and source description: Field 1 Sudan Silage

Sample date: 07/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	3,800.00	37,000.00		11.60
DL	0.01	0.01	0.01		1.00

Field 1 Sudan Second Cut

Sample and source description: Field 1 Sudan Second Cut

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	3,300.00	39,600.00		15.80
DL	0.01	0.01	0.01		1.00

2 - 10/03/2022: Oats, silage-soft dough

2 - 10/03/2022: Oats, silage-soft dough

Field 2 Oat Silage

Sample and source description: Field 2 Oat Silage

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,400.00	22,100.00		7.10
DL	0.01	0.01	0.01		1.00

2 - 05/20/2023: Sudangrass, silage

Field 2 Sudan Silage

Sample and source description: Field 2 Sudan Silage

Sample date: 07/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	3,700.00	37,200.00		11.80
DL	0.01	0.01	0.01		1.00

Field 2 Sudan Silage Second Cut

Sample and source description: Field 2 Sudan Silage Second Cut

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	3,200.00	39,000.00		16.10
DL	0.01	0.01	0.01		1.00

3 - 10/03/2022: Oats, silage-soft dough

3 - 10/03/2022: Oats, silage-soft dough

Field 3 Oat Silage

Sample and source description: Field 3 Oat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,400.00	3,300.00	26,100.00		9.00
DL	0.01	0.01	0.01		1.00

3 - 05/20/2023: Sudangrass, silage

Field 3 Sudan Silage

Sample and source description: Field 3 Sudan Silage

Sample date: 07/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	3,100.00	42,100.00		13.20
DL	0.01	0.01	0.01		1.00

Field 3 Sudan Silage Second Cut

Sample and source description: Field 3 Sudan Silage Second Cut

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	4,000.00	43,500.00		17.10
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

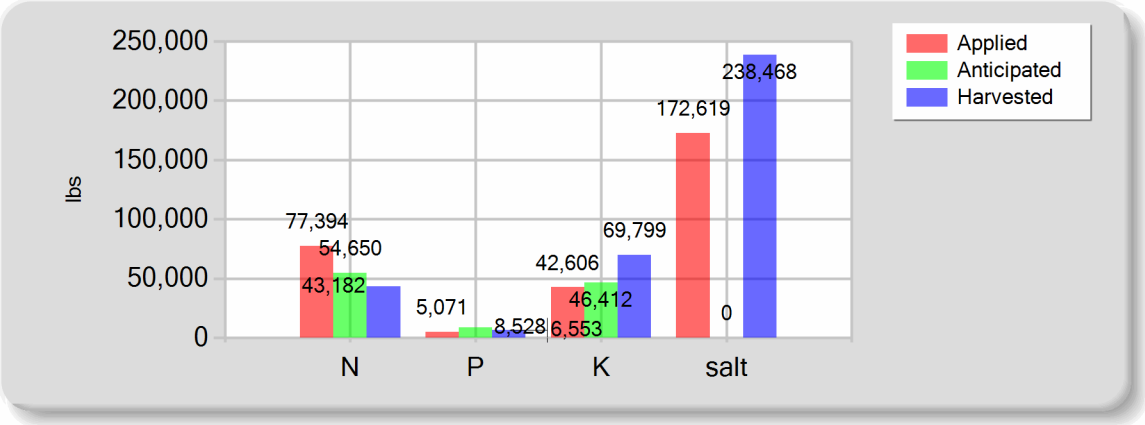
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

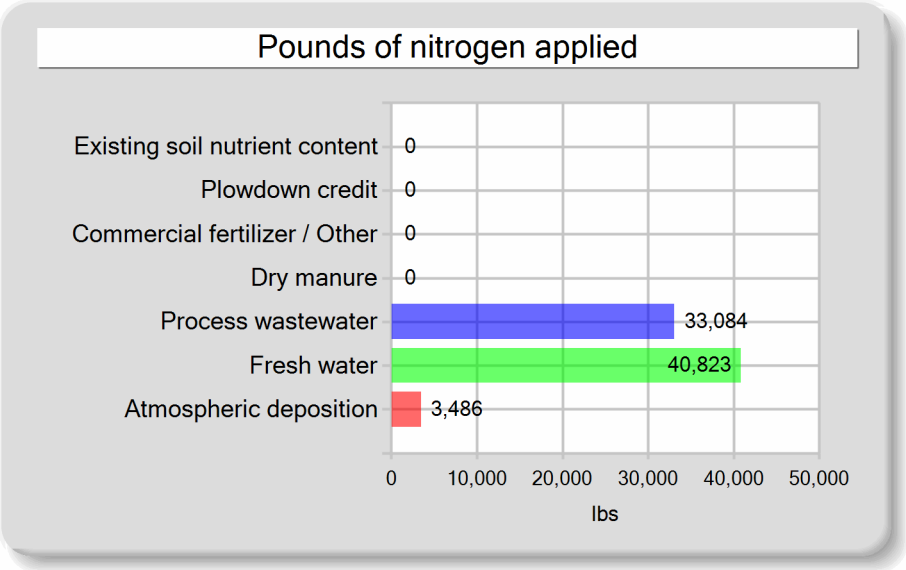
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

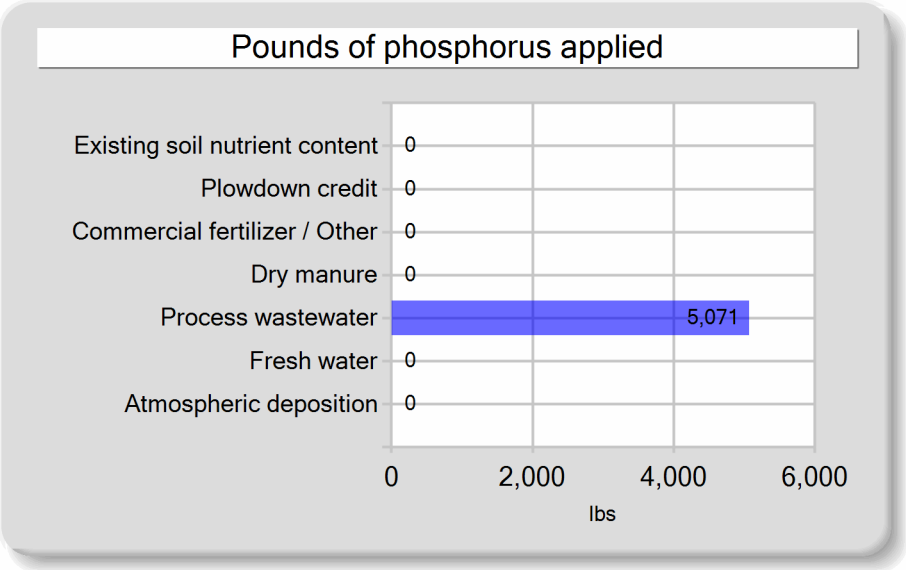
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	33,084.18	5,071.05	42,605.94	172,619.45
Fresh water	40,823.36	0.00	0.00	0.00
Atmospheric deposition	3,486.00	0.00	0.00	0.00
Total nutrients applied	77,393.54	5,071.05	42,605.94	172,619.45
Anticipated crop nutrient removal	54,650.00	8,528.00	46,412.00	0.00
Actual crop nutrient removal	43,181.59	6,553.09	69,799.18	238,468.23
Nutrient balance	34,211.96	-1,482.03	-27,193.24	-65,848.78
Applied to removed ratio	1.79	0.77	0.61	0.72

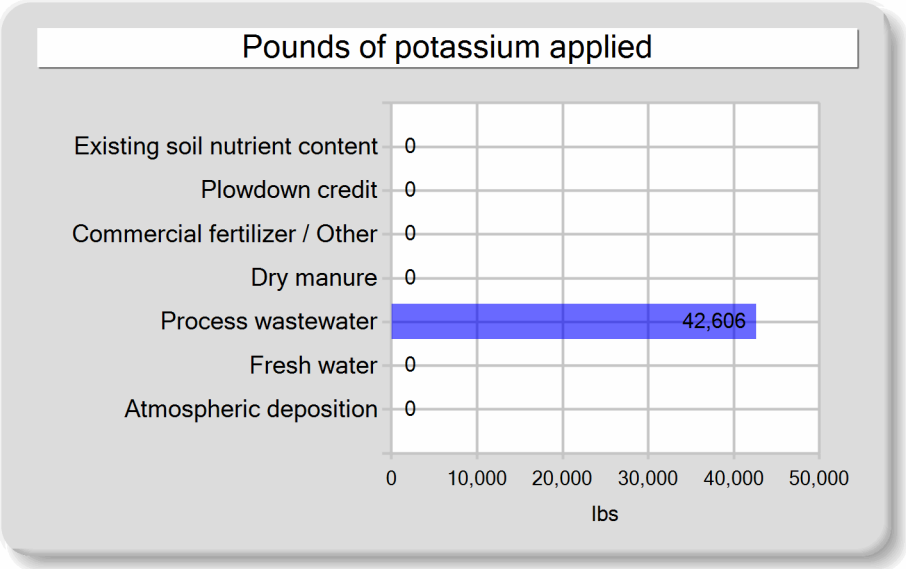
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

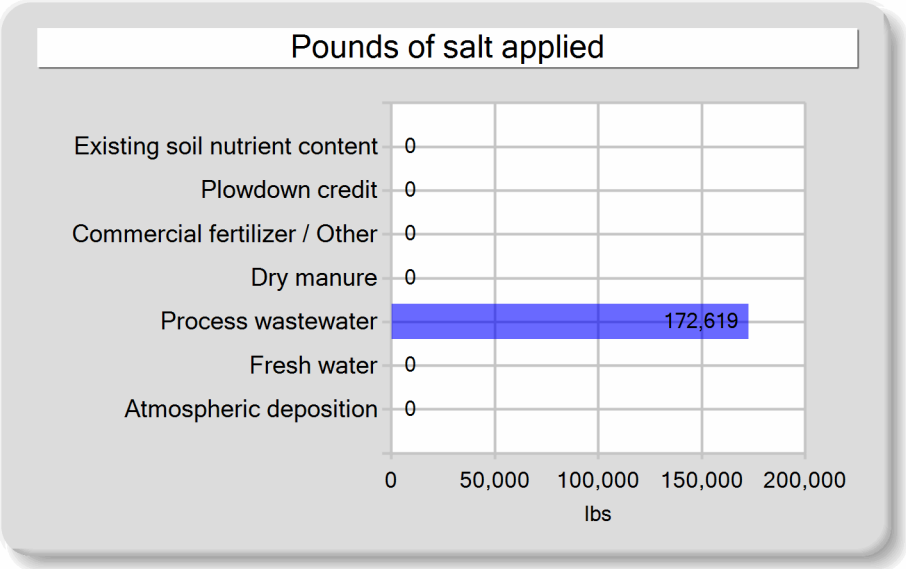


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 6 almonds removal above ratio, however yields reported are meat weights and not field weights

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Joe Alves

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe Alves	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Alves & Son Dairy #1
13745 Newport Rd.
Turlock CA 95303
00-0023570 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23K0515
Sampled By: M Dekeyrel
Requested By: Melissa Alves
Submitted Date: 11/09/23
Reported Date: 11/16/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	Irrigation Well	11/09/23	12:10	837	0.84	13.5	ND	*	7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Alves & Son Dairy #1
13745 Newport Rd.
Turlock CA 95303
00-0023570 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23K0515
Sampled By: M Dekeyrel
Requested By: Melissa Alves
Submitted Date: 11/09/23
Reported Date: 11/16/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	Irrigation Well	11/09/23	12:10	837	0.84	**13.5**	36.86	3.07

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #1
13745 Newport Rd.
Turlock, CA 95303

Account# 00-0023570
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:34

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0515-01	Irrigation Well	Ag Water	M Dekeyrel		11/09/2023 12:10

Default Cooler Temperature on Receipt °C: 0.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

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Turlock, CA 95303

Account# 00-0023570
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:34

Sample Results

Sample: Irrigation Well
23K0515-01 (Water)

Sampled: 11/9/2023 12:10
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.84**	mmhos/cm	0.01	1		11/15/23 11:36	SM 2510 B		BEK0350
Electrical Conductivity umhos	**837**	umhos/cm	10.0	1		11/15/23 11:36	SM 2510 B		BEK0350
Ammonia (as N)	ND	mg/L	0.00	1		11/09/23 12:10	Field		BEK0315
Nitrate Nitrogen as NO3N	**13.5**	mg/L	0.1	1	10	11/10/23 16:13	EPA 300.0		BEK0340
Temperature	**25.0**	units	0.0	1		11/15/23 11:36	SM 4500-H+	H	BEK0350
pH	**7.6**	units	1.0	1		11/15/23 11:36	SM 4500-H+	H	BEK0350

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LABORATORY INC

Alves & Son Dairy #1
13745 Newport Rd.
Turlock, CA 95303

Account# 00-0023570
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:34

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0340										
Blank (BEK0340-BLK1)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0340-BLK2)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0340-BLK3)					Prepared & Analyzed: 11/11/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0340-BS1)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEK0340-BS2)					Prepared & Analyzed: 11/11/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEK0340-DUP1)		**Source: 23K0518-01**			Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	11.6		0.1	mg/L		11.6			0.423	10
Duplicate (BEK0340-DUP2)		**Source: 23K0518-04**			Prepared & Analyzed: 11/11/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L		10.0			3.40	10
Matrix Spike (BEK0340-MS1)		**Source: 23K0518-01**			Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	16.9		0.1	mg/L	5.000	11.6	106	90-110		
Matrix Spike (BEK0340-MS2)		**Source: 23K0518-04**			Prepared & Analyzed: 11/11/2023					
Nitrate Nitrogen as NO3N	15.9		0.1	mg/L	5.000	10.0	117	90-110		
Reference (BEK0340-SRM1)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0340-SRM2)					Prepared & Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0340-SRM3)					Prepared & Analyzed: 11/11/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

Alves & Son Dairy #1
13745 Newport Rd.
Turlock, CA 95303

Account# 00-0023570
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:34

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350										
Blank (BEK0350-BLK1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.6		1.0	units						
Blank (BEK0350-BLK2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEK0350-BLK3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEK0350-DUP1)		**Source: 23K0483-01**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	3.23		0.01	mmhos/cm		3.27			1.28	10
pH	7.4		1.0	units		7.4			0.269	10
Electrical Conductivity umhos	3230		10.0	umhos/cm		3270			1.28	10
Duplicate (BEK0350-DUP2)		**Source: 23K0518-04**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	0.68		0.01	mmhos/cm		0.68			0.337	10
Electrical Conductivity umhos	685		10.0	umhos/cm		682			0.337	10
pH	7.8		1.0	units		7.8			0.256	10
Reference (BEK0350-SRM1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	576			umhos/cm	538.0		107	90-110		
Reference (BEK0350-SRM2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0350-SRM3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM4)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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DELLAVALLE™ LABORATORY INC

Alves & Son Dairy #1
13745 Newport Rd.
Turlock, CA 95303

Account# 00-0023570
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 11/09/2023 16:20
Reported: 11/16/2023 11:34

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350 (Continued)										
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEK0350-SRM6)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM7)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM8)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/09/23 16:20 23K0515

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Acct No. 11-9-23 Loc.

Bill To: 73570 8

Purchase Order No. Results Needed By

No. of Samples 1 No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Client **Alves & Son Dairy #1**
Address 13745 Newport Road
City, State, Zip Turlock CA 95380
Phone Fax
Cell/Email

Copy to Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by Melissa Alves - (209) 585-8221

Ranch

Date sampled 11/9/23

Sampled by MDelaurel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. Irrigation Well	Sampled From:	11/9/23	1210p	Ø	0.1
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MDelaurel	CAS Inc	11/9/23 1210p	11/9/23 1pm
Second	JLF	DLV	11-9-23 1:00	11-9-23 3:55
Third				
Fourth			11/9 16:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid Rec By Check No. Date

Shipping

$ In
$ Out

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

11/09/23 16:20 23K0515

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0446
Sampled By: M Dekeyrel
Requested By: Melissa Alves
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project:
Crop ID:

Alves & Son Dairy #1
13745 Newport Rd.
Turlock CA 95303
00-0023570 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - Silage	12/06/23	11:10	235	0.24	3.9	ND		8.1
2	DW - MB	12/06/23	11:10	962	0.96	8.7	ND		7.5

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Alves & Son Dairy #1
13745 Newport Rd.
Turlock CA 95303
00-0023570 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0446
Sampled By: M Dekeyrel
Requested By: Melissa Alves
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - Silage	12/06/23	11:10	235	0.24	**3.9**	10.65	0.89
2	DW - MB	12/06/23	11:10	**962**	**0.96**	**8.7**	23.75	1.98

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Alves & Son Dairy #1
13745 Newport Rd.
Turlock, CA 95303

Account# 00-0023570
Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 12/07/2023 15:35
Reported: 12/13/2023 11:17

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0446-01	DW - Silage	Ag Water	M Dekeyrel		12/06/2023 11:10
23L0446-02	DW - MB	Ag Water	M Dekeyrel		12/06/2023 11:10

Default Cooler Temperature on Receipt °C: -2.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Submitted By: Melissa Alves

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Reported: 12/13/2023 11:17

Sample Results

Sample: DW - Silage
23L0446-01 (Water)

Sampled: 12/6/2023 11:10
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.24**	mmhos/cm	0.01	1		12/11/23 16:03	SM 2510 B		BEL0301
Electrical Conductivity umhos	**235**	umhos/cm	10.0	1		12/11/23 16:03	SM 2510 B		BEL0301
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 11:10	Field		BEL0462
Nitrate Nitrogen as NO3N	**3.9**	mg/L	0.1	1	10	12/08/23 05:13	EPA 300.0		BEL0227
Temperature	**25.0**	units	0.0	1		12/11/23 16:03	SM 4500-H+	H	BEL0301
pH	**8.1**	units	1.0	1		12/11/23 16:03	SM 4500-H+	H	BEL0301

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Received: 12/07/2023 15:35
Reported: 12/13/2023 11:17

Sample Results
(Continued)

Sample: DW - MB
23L0446-02 (Water)

Sampled: 12/6/2023 11:10
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.96**	mmhos/cm	0.01	1		12/11/23 16:04	SM 2510 B		BEL0301
Electrical Conductivity umhos	**962**	umhos/cm	10.0	1		12/11/23 16:04	SM 2510 B		BEL0301
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 11:10	Field		BEL0462
Nitrate Nitrogen as NO3N	**8.7**	mg/L	0.1	1	10	12/08/23 05:33	EPA 300.0		BEL0227
Temperature	**25.0**	units	0.0	1		12/11/23 16:04	SM 4500-H+	H	BEL0301
pH	**7.5**	units	1.0	1		12/11/23 16:04	SM 4500-H+	H	BEL0301

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Received: 12/07/2023 15:35
Reported: 12/13/2023 11:17

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0227										
Blank (BEL0227-BLK1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK4)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0227-BS1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		106	90-110		
LCS (BEL0227-BS2)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0227-BS3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
Duplicate (BEL0227-DUP1)		**Source: 23L0251-01**			Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	2.6		0.1	mg/L		2.6			0.610	10
Duplicate (BEL0227-DUP2)		**Source: 23L0351-02**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			0.00	10
Duplicate (BEL0227-DUP3)		**Source: 23L0446-01**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	3.9		0.1	mg/L		3.9			0.00	10
Matrix Spike (BEL0227-MS1)		**Source: 23L0251-01**			Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	7.8		0.1	mg/L	5.000	2.6	103	90-110		
Matrix Spike (BEL0227-MS2)		**Source: 23L0351-02**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.04	105	90-110		
Matrix Spike (BEL0227-MS3)		**Source: 23L0446-01**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	9.1		0.1	mg/L	5.000	3.9	104	90-110		
Reference (BEL0227-SRM1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0227-SRM2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0227 (Continued)										
Reference (BEL0227-SRM3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0227-SRM4)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0301										
Blank (BEL0301-BLK1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0301-BLK2)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0301-BLK3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.3		1.0	units						
Duplicate (BEL0301-DUP1)		**Source: 23L0451-01**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.48		0.01	mmhos/cm		0.48			0.0421	10
pH	7.9		1.0	units		7.9			0.381	10
Electrical Conductivity umhos	476		10.0	umhos/cm		475			0.0421	10
Duplicate (BEL0301-DUP2)		**Source: 23L0453-02**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.80		0.01	mmhos/cm		0.79			0.126	10
Electrical Conductivity umhos	795		10.0	umhos/cm		794			0.126	10
pH	7.8		1.0	units		7.8			0.129	10
Reference (BEL0301-SRM1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	437			umhos/cm	426.0		103	90-110		
Reference (BEL0301-SRM2)					Prepared & Analyzed: 12/11/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0301-SRM3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0301-SRM4)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0301-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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Account Manager: Ben Nydam
Submitted By: Melissa Alves

Received: 12/07/2023 15:35
Reported: 12/13/2023 11:17

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0301 (Continued)										
Reference (BEL0301-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0301-SRM6)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0301-SRM7)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0301-SRM8)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		99.8	97.5-102.5		

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